

Dear Editors,

We are pleased to submit our manuscript entitled "*Critical Fragility Emerges from Chromosomal Instability in Cancer*" for consideration as a Research Article in **PLOS Computational Biology**.

Background and motivation. Chromosomal instability (CIN) is a recognized hallmark of cancer that plays a fundamental role in tumor evolution: while it accelerates adaptation and generates the karyotypic diversity required for progression, it simultaneously imposes a severe fitness cost by increasing the accumulation of deleterious genetic alterations. A central, unresolved question is whether this continuous instability inevitably pushes tumors to a predictable "edge of viability", a critical boundary where the burden of mutations threatens their own survival.

Model and Results. In this manuscript, we present a novel in silico framework that explicitly models diploid cell populations, karyotypic structural instability (polyploidy and aneuploidy), and discrete functional gene classes (oncogenes, tumor suppressors, housekeeping genes, and stability regulators) to bridge the microscopic rules of cell division with macroscopic tumor spatial dynamics. Using a combination of stochastic lattice simulations and mean-field analytical approximations, we demonstrate that cancer populations consistently self-organize toward a critical instability threshold, an "edge of viability." Below this boundary, instability fuels malignant expansion; beyond it, the population experiences mutational meltdown.

Impact. Furthermore, our model provides a quantitative blueprint for evaluating evolutionary therapeutic strategies. We show that interventions aimed merely at mass reduction are often ineffective. In contrast, treatments designed to push the system across this critical transition boundary—either by globally elevating genomic damage or by selectively eliminating low-instability reservoir clones—can successfully drive the tumor into an irreversible population collapse.

Relevance for PLOS Computational Biology's readership. We believe this work is highly aligned with the scope of PLOS Computational Biology. By formalizing cancer progression as a runaway evolutionary process approaching a phase transition, we provide a mathematically grounded, mechanistic understanding of how complex intracellular genomic dynamics govern tissue-level tumor fate. Our findings reframe chromosomal instability not merely as a driver of cancer, but as a latent, targetable vulnerability.

For the above reasons, we are confident that you will consider our study for potential publication in *PLOS Computational Biology*.

This manuscript has not been published and is not under consideration for publication elsewhere. All authors have read and approved the manuscript and consent to its submission to PLOS Computational Biology.

Thank you for your time and consideration of our work. We look forward to your response.

Sincerely,

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